



Stands for Painting Cylindrical Components

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

Borusan Makina Kazakhstan has developed stands to improve the painting process of cylindrical components such as hydraulic cylinders, axles and many others.

In previous applications, these components were painted using pallets or wooden boxes. The process involved the use of a crane or forklift to turn the cylindrical components over and consequently required a lot of effort from the painter.

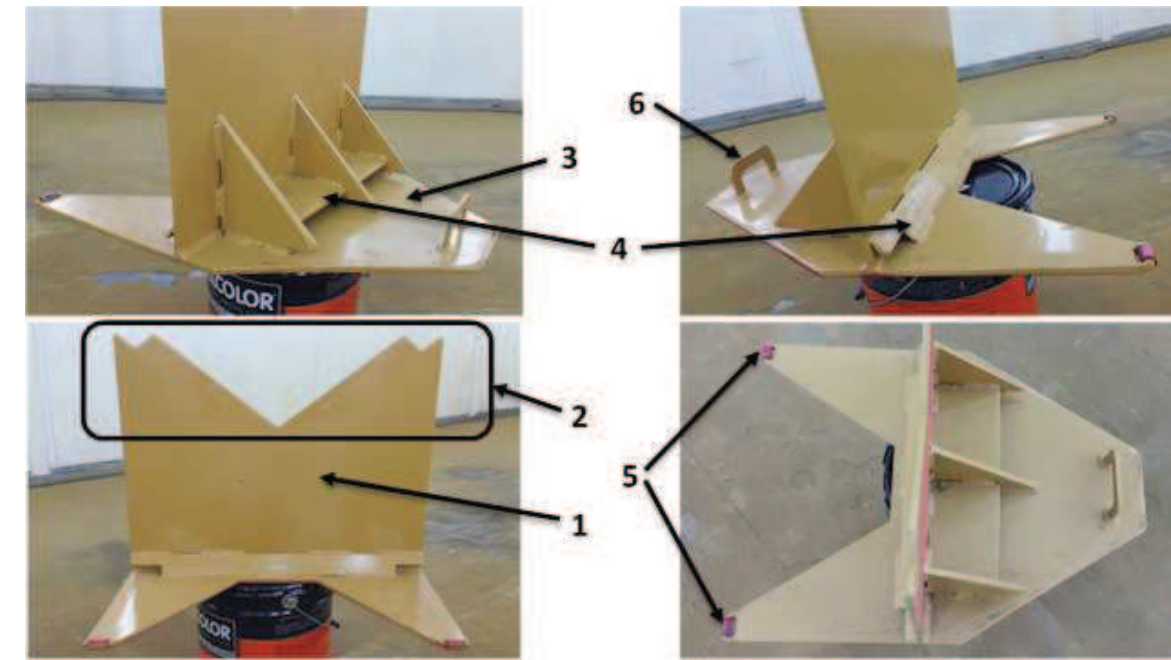
The usage of stands offsets the need for a forklift or crane while allowing the painter to roll and paint the components easily and efficiently. Moreover, it significantly decreases the preparation time required for painting applications since the stands are arranged according to the size of the cylindrical components. As a result, these stands become very simple to use and very easy store.

The stands serve purpose in multiple tasks such as washing and painting cylindrical components, repairing and temporarily storing hydraulic cylinders and axles, as well as maintaining crankshafts. In other areas, it is necessary to use soft pads on the stands to avoid friction on the metal surfaces.

2.0 Best Practice Description

One set consists of two stands. Each stand is made from steel sheets with a thickness of at least 10 mm. They are designed to be reliable for the installation of cylindrical components (axles, hydraulic cylinders, etc.).

Each stand consists of (1) one support with (2) three angled working surfaces, (3) one foot with (4) one reinforcing plate on each side, (5) two wheeled rollers and (6) one Handle.



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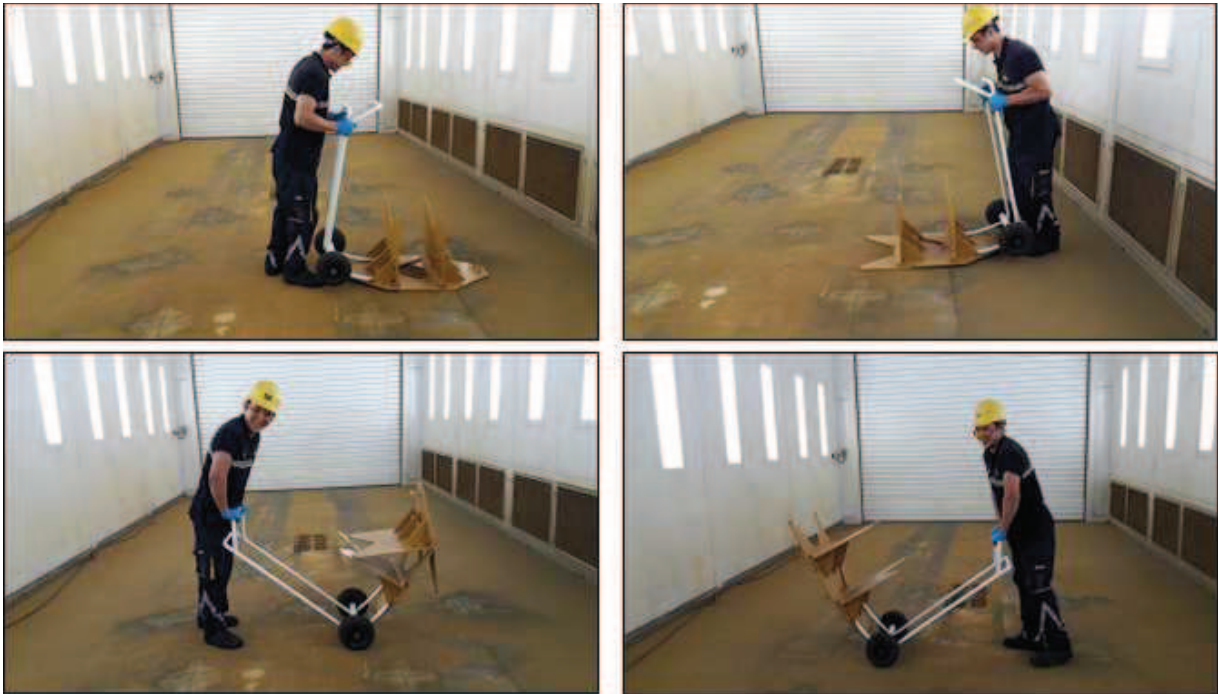
The shape of the working surface of two stands (each stand has three internal angles of 100⁰ each) allows the installation of either one big component or two small components on them.

The weight of each of the stands exceeds 20 kg which makes it difficult to move them around the shop. This explains why a pushcart is applied to ensure a safe displacement. It is manufactured with a 26-mm diameter bar and couple 240-mm diameter wheels.

This pushcart is practical for taking a set of stands from the storage to the working area and arranging them with respect to the size of the component ready to be painted. Following the task completion, the stands are easily stored back using the pushcart. Out of the two stands in a set, only one can be moved manually because of the handle and rollers it possesses.



Because of the design of the stands, it is possible to join two stands from both sides and transport them with the pushcart.



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When following the attached drawing the maximum load is 1 ton. In addition, we can identify load depend on the diameter of the stand - do not put components that exceed this diameter.

3.0 Implementation Steps

The stands and pushcart are in use.

- 1) Take the stands from the storage to the working area with the pushcart and arrange them with respect to the size of the component.



- 2) Install the component for painting onto the stands. Note that in certain cases, when the axles are installed onto the stand, they might need a support under the shank.



Note: Working Load Limit: WLL 1 Ton. Do not exceed the limit to avoid breaking the stands.

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- 3) Paint the component. Since the shape of the working edge is designed to minimize frictional contact between the stand and the component, it leaves a small unpainted trace representing only 0,5% of the whole component's surface.



- 4) After the component is fully painted, displace it from the stands and transport it to the dispatch area. Next, move the stands (if they are not needed for another component) to the storage by means of a pushcart.



The stands are marked with signs for safety purposes.



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4.0 Benefits

- Easy and convenient to use;
- Easy and convenient to store;
- Reduced component preparation time for painting (the stands can be easily arranged as per the size of the component);
- Easy to turn painted components over;
- Multipurposed components that can be used for washing and painting of cylindrical components; for repairing and temporarily storing hydraulic cylinders and axles, as well as maintaining crankshafts, etc.

5.0 Resources Required

- Materials needed to fabricate stands and pushcart;
- Welding.

6.0 Supporting Attachments / References

The necessary drawings of the stand and pushcart are attached.

7.0 Related Best Practices

None.

8.0 Acknowledgements

The idea of creating & implementing this Best Practice belongs to:

Sergey INYUTIN
Painter
Equipment Rebuild Shop
Borusan Makina Kazakhstan LLP

Azamat ORAZALINOV
Electrical & Gas Welder
Equipment Rebuild Shop
Borusan Makina Kazakhstan LLP

Bauyrzhan AKHANOV
Design Engineer
Borusan Makina Kazakhstan LLP
bakhanov@borusan.com

This Best Practice was written by:
Gulnaz MANSUROVA
CRC Process Engineer
Borusan Makina Kazakhstan LLP
gmansurova@borusan.com
+7 (701) 951 98 14

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